Intubation and transfer of suspected COVID-19 patients requiring mechanical ventilation in Isolation Room (ED)

Priorities:

1. **Plan, Prepare, Communicate.**
2. **Minimise exposure to staff and equipment.**
3. **Maintain staff protection.**

1. Referral of patient to intubate in isolation room A&E	
Referral pathway	Emergency: 550 ITU on call, who then liaises with ITU consultant on call Non emergency: direct referral to ICU Consultant on call, who delegates available staff (anaesthetist and ODP) to send to A&E
2. Staffing required	
Inside isolation room - intubation team	- Anaesthetist (full PPE with FFP3 mask) (Consultant or senior trainee) - ODP (full PPE with FFP3 mask) – **anaesthetist to bleep 017 to arrange.** - Do not use obstetric anaesthetist/ODP
In anteroom	- 1 nurse (full PPE), in red zone of anteroom, ready to hand over any extra equipment, as required. Also to help transfer to HDU afterwards. - 1 runner, surgical mask and apron, ready to leave room to get further help / equipment + responsible for scribing
3. Equipment to *bring* to Emergency department (by the anaesthetist)	
Intubation tray	Ready-made intubation trays, **stored on ICU** + 2 available in **cupboard of anteroom** of A&E isolation room (made up by ICU nurses): - Mac 4 blade and handle (PROACT) in sealed bag - Size 7 and 8 ETT - Tube tie - Catheter mount and closed suction kit - HME filter - In-line EtCO2 - 20 ml syringe - Sterile gel sachet - Tube clamp - Magills forceps - RSI checklist

RSI Drugs	Ready-made in disposable tray : made fresh at the start of each morning shift on ICU by ICU resident (all syringes capped with red bungs): - Rocuronium 1 syringe of 100 mg (10 ml) - 2 extra rocuronium ampoules and a drawing up needle - Metaraminol 0.5 mg/ml - 20 ml syringe and 200 mg Propofol ampoule (for sedation after intubation, to be drawn up as needed) - 20ml syringe labelled Ketamine and drawing up needle. Ketamine to be requested from ED and drawn up when needed - Ketamine 10 mg/ml 20 ml : draw up neat - Ketamine 50 mg/ml 10 ml : draw up 4 ml = 200 mg, dilute with 16 ml NaCl 0.9% to 20 ml total, concentration = 10 mg/ml

4. Equipment which is present in isolation room :	
Videolaryngoscope	C-MAC confined to isolation room (One single use blade to be brought in at time of intubation, according to need)
Trolley by bedside	Drawer A: size 6.0 ETT Drawer B: size 3, 4 and 5 Ambu AuraGain supraglottic airways Drawer C: green, red and orange Guedel ; yellow, green and orange face mask Drawer D: scalpel and size 6.0 ETT in FONA pack, CICO pack Bottom drawer : Easy-IO kit and IO needles, minimal consumables for hygiene (Ex inco pads) On top : - Hamilton transport ventilator with single use breathing circuit - Defibrillator with adult and paediatric pads - 1 bag valve mask - 1 closed suction kit - 1 stethoscope (not to be used if possible)
Wall cupboard	- Empty. Do not open. Cardiac Arrest drugs box on top of cupboard (Adrenaline 1 in 10,000 1 mg in 10 ml PFS x 6, Amiodarone 300 mg in 10 ml PFS x1, Calcium Chloride 10% 10 ml PFS x 1)
Other	- Monitor and transport monitor - 2 pumps - Wall-suction and portable suction - O2 cylinder (small, ensure sufficiently filled for transfer) - 1 small procedure trolley - Phone and list of phone numbers on the wall - 1 orange bins - 1 sharps bin
Side room of isolation room (toilet)	- Ultrasound machine

4. Equipment which is present in isolation room :	
Anteroom of isolation room	- 2 orange bins - 1 trolley with all PPE equipment - 1 cupboard with consumables (syringes, canulation packs, blood bottles, blood culture bottles, swabs, patient gown, vomit bowls) - Procedure posters on wall (out-of-roo m intubation checklist, donning and doffing procedure)
Outside isolation room in Resus	! Plan ahead and only take the items that you need inside isolation room - Difficult Airway trolley and Paediatric Critical Care trolleys. Trolley to go in to room in a declared emergency only - NIV machine
5. Preparation prior to entering isolation room	
Ensure all equipment available (checklist available in anteroom)	- Follow the COVID Outside Room Checklist : - **Minimise exposure to staff and equipment** - Decide who needs to go in (Maximum 3) and what equipment you will definitely need - Intubation tray (confirm plan A -> D prior to donning PPE) - RSI drugs (confirm doses prior to donning PPE) - Anaesthetic facemask of appropriate size - C-Mac blade (size 3, 4 or D blade), Stylet if using D blade - Bougie - Water circuit with HME filter
Other	Remove Personal belongings (phones, badges, bleep) and place in clear plastic bag / blood sample bag. - to be kept secure by runner, and collected at the end

6. Intubation procedure	
Pre-intubation	- Set up ventilator and circuit with ETCO2 and HME ready to go - Check all connections are tight - Position C-Mac so anaesthetist and ODP can see - COVID-19 in-room RSI checklist, to be read out loud - Preoxygenate with 2 hand technique, tight seal
Post-intubation	- Read out COVID-19 post RSI and pre move checklist - Clean up any gross contamination (vomit and respiratory secretions) - Remove outer pair of gloves, wash inner pair of gloves and renew outer gloves
Other	- Move to Critical care area for placement of invasive lines, NG, etc - No CXR post intubation (Image after all lines and NG placed)

7. Pre departure to HDU/ICU

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Notify HDU / ICU	- Call receiving team; are they ready to accept? - Any syringes to be made up? (ex sedation, vasopressors) - Give estimated time of arrival to allow receiving staff to don PPE
Notify extra help	- Request 4 staff (porters / runners) to be outriders for transfer to critical care - Ensure A&E cleaning staff alerted that isolation room will require cleaning shortly
Notes	- Document retrospectively after you have transferred - Ensure runner has photocopied any ED notes
Patient preparation	- Portable monitor (ensure sufficiently charged), full monitoring during transfer including ETCO2 - Waters circuit - Tube Clamp - Supraglottic airway, Guedel - Transfer drugs – sedation, vasopressor, muscle relaxant - Portable suction - O2 cylinder 3/4 full - Transfer bag not required * Minimise cluttering of trolley, ensure nil hanging out from sides
Pre-departure decontamination of trolley	- Ensure all obvious contamination with bodily fluids on patient or trolley cleaned prior to moving in to ante room - Cover patient with theatre drape if available or sheet / blanket

8. Departure to HDU/ICU	
Leaving isolation room	- Anaesthetist, ODP and nurse keep their PPE on. - Runner and nurse in anteroom to leave the anteroom. Nurse to take bag of staff's personal belongings and patient notes. - Anaesthetist and ODP roll patient into anteroom. - Wipe down surfaces with Clenill wipes - Once isolation room door closed, door to hallway can be opened to transfer patient out into hallway - Anaesthetist and ODP to ensure they touch nothing in anteroom, and trolley not to touch any surfaces / walls
Path	- Shortest path - through radiology department - Use designated lift. (closest to theatres)

8. Departure to HDU/ICU	
Transfer	- 4 helpers minimum required to assist with clearing the pathway during transfer, 2 in the front, 2 in the back - Goal is for all staff / patients / visitors to be cleared of the pathway, and for anaesthetist / ODP accompanying patient not to have to touch any surfaces - Helpers to keep doors open, call lift, keep lift open, press button inside lift - Helpers to stay at safe distance from trolley at all times
Other	- Ensure isolation room immediately deep cleaned - Ensure any stock used inside the isolation room immediately refurnished after departure

9. Arrival HDU/ICU	
Staff required	- Minimum of 2 nursing staff from ICU/HDU - Transferring Anaesthetist, ODP - Assistant
Equipment required	- Prepare ICU ventilator settings - Blue single use transfer sheet - All ICU monitoring equipment available including ETCO2 - Pumps for sedation / vasopressors - Sedation +/- vasopressor syringes ready
Procedure	Before / after transfer of patient onto bed: - Clamp ETT (if possible at end of inspiration) - Stop transfer ventilator - Disconnect tubing and reconnect to ICU ventilator - Start ICU ventilator - Remove clamp
Invasive Lines	- Stay in PPE to place any invasive lines, ng tube, etc - Scrub and place sterile gown and sterile gloves over your PPE
Cleaning	- Ensure adequate decontamination of trolley and all transfer monitoring, ventilator, etc with Green Clinell wipes. - Single use equipment not used / no longer required to stay with the patient (do not dispose unless grossly contaminated)
Doffing	- Follow the PHE guide to doffing.